Dual-Stream Confidence Model — per-edge confidence and source-class metadata

SPEC-TH-040 v0.1 DRAFT Jeremie Nunez

1 Context

The Thalamus knowledge graph currently stores edges as untyped relations produced by the OSINT nano-swarm. In a dual-stream deployment (open-source intelligence fused with a high-trust field feed), edges produced from noisy open sources and edges produced from a tactical data-link, sensor-fusion bus, or mission debrief must be told apart at query time, at sweep time, and in any analyst-facing briefing. This spec defines the **confidence** scalar and **source_class** discriminator carried on every edge, the promotion and demotion rules that move an edge between classes, and the invariants that prevent an OSINT-only claim from ever masquerading as a field-confirmed fact.

2 Goals

- Every edge written to the knowledge graph carries a `confidence` $\in [0, 1]$ and a `source_class` $\in \{$`FIELD_HIGH`, `FIELD_LOW`, `OSINT_CORROBORATED`, `OSINT_UNCORROBORATED`$\}$.
- Promotion and demotion of an edge between classes is deterministic, auditable, and replayable from the edge history.
- Queries can filter and rank edges by source class and confidence without joining external tables.
- Sweep can target `OSINT_UNCORROBORATED` edges older than a configurable window for analyst review.
- The edge schema is backward compatible: existing edges are migrated to `OSINT_UNCORROBORATED` with `confidence = 0.3` unless a richer provenance is present.

3 Non-Goals

Non-Goal

This spec does *not* define the field-stream ingestion transport, the correlation algorithm, or the latency budget of the fusion loop. Those live in SPEC-TH-041 (Field-Correlation Cortex). It also does not define a probabilistic belief-propagation model across multi-hop paths — confidence is a per-edge scalar, not a graph-wide Bayesian posterior.

4 Invariants

Invariant

I-1 — Class floor. An edge whose only provenance is an open-source observation is never written with `source_class = FIELD_HIGH` or `FIELD_LOW`. OSINT provenance alone forbids a field class.

Invariant

I-2 — Promotion requires corroboration. An OSINT edge is promoted to `OSINT_CORROBORATED` only when at least one independent open source agrees within the corroboration window, *or* a field event matches its entity pair and relation. Promotion to

`FIELD_HIGH` requires a field event *and* a matching relation — never a pure OSINT signal.

Invariant

I-3 — Field dominance. A field event always overrides OSINT confidence on a matching edge. OSINT signals can never demote `FIELD_HIGH`; only a contradicting field event, an explicit analyst override, or expiry of the field freshness window can.

Invariant

I-4 — Bounded confidence. `confidence` is clamped to $[0, 1]$ at every write. Promotion and demotion apply monotonic transforms that respect the class ordering: `OSINT_UNCORROBORATED` $\leq$ `OSINT_CORROBORATED` $<$ `FIELD_LOW` $\leq$ `FIELD_HIGH` on the confidence axis within each class's allowed sub-range.

Invariant

I-5 — Auditability. Every confidence or source-class change emits an `EdgeProvenanceEvent` with `actor` (`osint-swarm`, `field-correlation`, `analyst`, `sweep`), `reason`, `previous`, and `next` values. No silent mutations.

5 Schema additions

```
// Extends the ResearchEntityEdge row in packages/db-schema/src/
   knowledge-graph/edges.ts

export type SourceClass =
  | "FIELD_HIGH"              // confirmed by a high-trust field event
  | "FIELD_LOW"               // field event present but partial / low-
     rate
  | "OSINT_CORROBORATED"      // >= 2 independent OSINT sources agree
  | "OSINT_UNCORROBORATED";   // single OSINT hypothesis, not yet
     confirmed

export interface EdgeConfidence {
  value: number;              // [0, 1]
  sourceClass: SourceClass;
  lastPromotedAt: Date | null;
  lastDemotedAt: Date | null;
  corroborationCount: number; // distinct sources agreeing
  fieldEventId: string | null;// link to field event if class in {FIELD_
     *}
}

export interface EdgeProvenanceEvent {
  edgeId: number;
  actor: "osint-swarm" | "field-correlation" | "analyst" | "sweep";
  reason: string;
  previous: Pick<EdgeConfidence, "value" | "sourceClass">;
  next: Pick<EdgeConfidence, "value" | "sourceClass">;
  at: Date;
}

export interface PromoteEdgeInput {
  edgeId: number;
  evidence:
```

```
    | { kind: "osint-corroboration"; sources: string[] }
    | { kind: "field-match"; fieldEventId: string; stream: string };
}

export interface DemoteEdgeInput {
  edgeId: number;
  evidence:
    | { kind: "field-contradiction"; fieldEventId: string }
    | { kind: "analyst-override"; analystId: number; note: string }
    | { kind: "field-freshness-expired" };
}
```

5.1 Confidence bands per class

source_class	Allowed confidence	Typical write-time value
OSINT_UNCORROBORATED	$[0.10, 0.50]$	0.30
OSINT_CORROBORATED	$[0.40, 0.75]$	0.55
FIELD_LOW	$[0.65, 0.85]$	0.75
FIELD_HIGH	$[0.85, 1.00]$	0.92

6 Promotion and demotion rules

- **OSINT swarm write** — new edge always starts at `OSINT_UNCORROBORATED`, `confidence = 0.30`, `corroborationCount = 1`.
- **OSINT corroboration** — a second independent source for the same `(subject, relation, object)` within the corroboration window promotes to `OSINT_CORROBORATED`, `confidence = 0.55`. Additional corroborating sources may raise `confidence` up to 0.75 asymptotically.
- **Field match** — a field event matching the edge promotes to `FIELD_LOW` (partial field evidence) or `FIELD_HIGH` (confirming field evidence on a critical stream), sets `fieldEventId`, `confidence` = the band's typical write-time value.
- **Field contradiction** — a field event contradicting the edge demotes to `OSINT_UNCORROBORATED`, `confidence = 0.15`, and flags the edge for analyst review via Sweep.
- **Field freshness expiry** — a `FIELD_*` edge whose backing field event is older than the freshness window degrades one class step, with `confidence` recomputed to the new band's upper bound.
- **Analyst override** — manual set of `confidence` and `source_class`, recorded as `analyst` actor. Cannot place an OSINT-only edge in a `FIELD_*` class (enforced by I-1).

7 Interface

```
// packages/thalamus/src/cortices/confidence.ts

export interface ConfidenceService {
  promote(input: PromoteEdgeInput): Promise<EdgeConfidence>;
  demote(input: DemoteEdgeInput): Promise<EdgeConfidence>;
  read(edgeId: number): Promise<EdgeConfidence>;
  history(edgeId: number): Promise<EdgeProvenanceEvent[]>;
  query(filter: {
    minConfidence?: number;
    sourceClasses?: SourceClass[];
    olderThan?: Date;
```

```
  }): Promise<Array<{ edgeId: number; confidence: EdgeConfidence }>>;
}
```

8 Scenarios

Scenario: OSINT edge starts uncorroborated

Given a cortex in the OSINT swarm emits a new edge for `(Actor A, associatedWith, Campaign C)`
When `ConfidenceService.promote` is not yet called
Then the edge is written with `source_class = OSINT_UNCORROBORATED`
And `confidence = 0.30`
And an `EdgeProvenanceEvent` is emitted with actor `osint-swarm`

Scenario: OSINT corroboration promotes confidence

Given the edge above exists at `confidence = 0.30`
When a second independent OSINT source emits the same triple within the corroboration window
Then the edge is promoted to `OSINT_CORROBORATED`
And `confidence = 0.55`
And `corroborationCount = 2`

Scenario: Field match promotes to FIELD_HIGH

Given an `OSINT_CORROBORATED` edge at `confidence = 0.55`
When a friendly-force tracking event matches the subject, relation, and object
Then the edge is promoted to `FIELD_HIGH`
And `confidence = 0.92`
And `fieldEventId` is populated
And the provenance history records the transition

Scenario: Invariant I-1 forbids OSINT-only FIELD_HIGH

Given an edge with only OSINT provenance
When a caller attempts `promote` with no `field-match` evidence
Then the call is rejected with `InvalidPromotion`
And the edge class and confidence are unchanged
And no `EdgeProvenanceEvent` is emitted

Scenario: Field contradiction demotes to OSINT_UNCORROBORATED

Given a `FIELD_LOW` edge at `confidence = 0.75`
When a contradicting field event arrives on the tactical data-link
Then the edge is demoted to `OSINT_UNCORROBORATED`
And `confidence = 0.15`
And the edge is flagged for Sweep analyst review

Scenario: Field freshness expiry

Given a `FIELD_HIGH` edge whose backing field event is older than the freshness window
When the freshness sweep runs

Then the edge is degraded one class step to `FIELD_LOW`
And `confidence` is recomputed to the upper bound of the `FIELD_LOW` band
And the provenance event records `field-freshness-expired`

9 Acceptance Criteria

Acceptance Criterion AC-1

Every edge insert produced by any OSINT cortex persists with `source_class = OSINT_UNCORROBORATED` and `confidence` in $[0.10, 0.50]$.

Acceptance Criterion AC-2

A second independent OSINT source on an existing edge promotes it to `OSINT_CORROBORATED` and records an `EdgeProvenanceEvent`.

Acceptance Criterion AC-3

`ConfidenceService.promote` with `field-match` evidence moves the edge to `FIELD_HIGH` (or `FIELD_LOW` per stream policy) and sets `fieldEventId`.

Acceptance Criterion AC-4

`ConfidenceService.promote` without `field-match` evidence never produces a `FIELD_*` class. The call fails with a typed error.

Acceptance Criterion AC-5

A contradicting field event demotes a `FIELD_*` edge to `OSINT_UNCORROBORATED` with `confidence` ≤ 0.20.

Acceptance Criterion AC-6

No OSINT actor can demote a `FIELD_HIGH` edge. Attempts are rejected and logged.

Acceptance Criterion AC-7

`ConfidenceService.history(edgeId)` returns every class or confidence transition in insertion order. No gaps.

Acceptance Criterion AC-8

`ConfidenceService.query` can return all `OSINT_UNCORROBORATED` edges older than a caller-supplied timestamp, for Sweep to pick up.

10 Traceability

AC	Test file (planned)	Test name
AC-1	`packages/thalamus/tests/field-correlation/confidence-write.spec.ts`	`OSINT writes`
AC-2	`packages/thalamus/tests/field-correlation/confidence-promote.spec.ts`	`second OSINT`
AC-3	`packages/thalamus/tests/field-correlation/confidence-promote.spec.ts`	`field match p`
AC-4	`packages/thalamus/tests/field-correlation/confidence-invariants.spec.ts`	`OSINT-only ca`
AC-5	`packages/thalamus/tests/field-correlation/confidence-demote.spec.ts`	`field contrad`
AC-6	`packages/thalamus/tests/field-correlation/confidence-invariants.spec.ts`	`OSINT cannot`
AC-7	`packages/thalamus/tests/field-correlation/confidence-history.spec.ts`	`history is co`
AC-8	`packages/thalamus/tests/field-correlation/confidence-query.spec.ts`	`sweep-targeta`

11 Open Questions

- Should `FIELD_LOW` vs `FIELD_HIGH` be a property of the stream, of the event, or of the correlation score? Current draft: stream policy sets the default, per-event override allowed.
- Freshness window per stream: a single global TTL, or per-stream configurable? Leaning per-stream.
- Do we persist provenance events in Postgres (auditable SQL) or ship them to the observability bus only? Current draft: both — SQL for audit, bus for dashboards.
- Multi-hop confidence: out of scope for this spec but should be tracked in a follow-up.